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Motor Imagery Classification by Means of Source Analysis Methods

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Abstract — **We report our investigation of classification of imagined left and right hand movements by applying source analysis methods. Independent component analysis (ICA) is used as a spatio-temporal filter, then equivalent dipole analysis and cortical current density imaging methods are applied to reconstruct equivalent sources, to aid classification of motor imagery tasks in a human subject. The classification was considered correct if the equivalent source was found over the motor cortex in the corresponding hemisphere. A classification rate of about 80% was achieved in the human subject studied using both the equivalent dipole analysis and the cortical current density imaging analysis.**
Keywords — **Brain Computer Interface, Source Analysis, Inverse Solution, EEG, Independent component analysis**

I. Introduction

Brain Computer Interface (BCI) provides a novel communication channel between the human brain and computer [1-9]. It helps patients suffering from severe physical disabilities (locked-in) but being cognitively intact to interact with environment through decoding their mental states. An important step in BCI is the classification of EEG signals recorded from the scalp during different kinds of motor imagery such as left or right hand movement imagination. In the previous work, different features are extracted in frequency (i.e. *mu* rhythm, beta rhythm), time (i.e. ERD and ERS) and spatial domain (i.e. spatial pattern correlation), and then these features are translated and identified by using either nonlinear classifiers (i.e. Artificial Neural Network, etc) or linear classifiers (i.e. Linear Discriminant Analysis, etc) [1-9]. In these methods, both training and testing groups are needed -- using the best features extracted from the training group to classify the signals in the testing group. In the present study, we test the hypothesis that the source analysis approach will aid the classification of motor imagery, thus facilitating BCI from scalp EEG.

II. Methods

1) Data Description

The EEG data files consisting of the synchronized imaginary movement experiments were made available by Dr. Allen Osman of University of Pennsylvania [3] [4].

Scalp EEG data were recorded from 59 electrodes placed according to the International 10/20 System with sampling rate of 100 Hz. The subjects were asked to imagine either right or left hand movement (180 trials, 90 left and 90 right) according to a highly predictable time cue.

Each trial (6 second) began with a blank screen, Two important timing cues that should be mentioned here are that at 3.75 seconds of a trial epoch, a cue (*preparation cue*, lasting 250 ms) of one letter ("L" or "R") appeared on the screen indicating which hand movement should be imagined; and at 5.0 seconds another cue 'X' (*execution cue*, lasting 50 ms) appeared, indicating that it was time to make the requested response as is shown in Fig. 1.

As described below, for each trial, the time period from 4.5-5.0s was selected. And the electrodes chosen for source reconstruction were FC3, FC1, FCz, FC2, FC4, C3, C1, Cz, C2, C4, CP3, CP1, CPz, CP2, CP4.

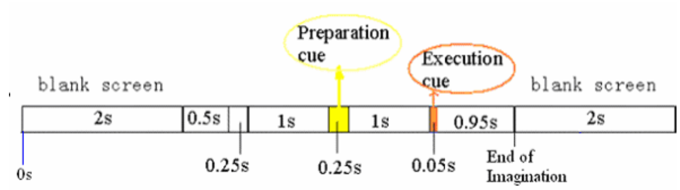

Fig. 1 Time sequence of one trial epoch of the experiment

2) Spatial Filtering

Surface Laplacian filter is a sort of spatial high pass filter, it attempts to enhance neuronal activity which is close to the recording electrode [10]. It could be approximately calculated by using the five-point approximation method [11].

$$V_j^{Lap} = V_j - \frac{1}{4}\sum_{k \in S_j} V_k \qquad (1)$$

where V_j is the scalp potential EEG of the j-th channel, and S_j is an index set of the four neighboring channels.

3) Band Pass Filtering

The preprocessed EEG data were band-pass filtered in the alpha band (11-12Hz), as it was reported that during motor imagery synchronization in alpha band sometimes emerges over the ipsilateral central region from the start of the movement imagination [12].

4) Independent Components Analysis (ICA)

The ICA method is to extract independent sources signals from their linear mixtures. The goal is to find a coordinate system in which the data are maximally statistically independent, not merely decorrelated:

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$$M_{n\times t} = A_{n\times n} \cdot S_{n\times t} \quad (2)$$

where M is the original data matrix of n (channel number) by t (time points). A is the mixing matrix and S is a matrix containing independent components.

Solving the ICA problem is to find a matrix W iteratively so that the linear transformation of the original data M yields components that are as mutually independent as possible.

$$\hat{S} = \hat{W} \cdot M \quad (3)$$

Before implementing ICA, Singular value decomposition (SVD) is used for decorrelation. This procedure can speed up the iteration process of ICA by setting all singular values to zero which are below a certain threshold. It is realized as

$$M = U \cdot \Sigma \cdot V^T \quad (4)$$

where Σ is a diagonal matrix containing singular values of M. U contains the orthogonal, normalized spatial patterns and V^T contains the normalized time courses. Assuming that V_c is the subset of time courses from V^T corresponding to singular values above threshold, which is regarded as signal subspace, ICA can be computed and realized only on this signal subspace. Then, equation (3) can be changed to

$$\tilde{S} = \tilde{W} \cdot V_c \quad (5)$$

Replacing V^T in equation (4) by V_c from equation (5), the mixing matrix A in equation (2) can be obtained by following equation

$$A = U\Sigma\widetilde{W}^{-1} \quad (6)$$

Sorting mixing matrix A by its column norm, independent components can be obtained from equation (2) in non-decreasing order.

Because we are interested in the activity in motor cortex and only the electrodes around motor cortex are chosen for source reconstruction, only the first component is chosen into the data reconstruction. Then the time point with the largest amplitude in the first component of ICA is used for source reconstruction.

5) Source Analysis

We have used the equivalent dipole analysis and cortical current density (CCD) imaging approaches in the present study. The 3-concentric-sphere head model was used to approximate the head volume conductor.

In the equivalent dipole model, to find the location and moment of the dipole is to minimize the residual error:

$$\Delta^2 = \left\| Lj - m \right\|^2 \quad (7)$$

where L is the lead field matrix, j is the dipole moment and m is the measured potential. The simplex method was used to find the location and moment of the equivalent dipole.

CCD is defined as dipole moment per volume with the unit: (μAmm/mm^3). A current density distribution is discretized into a large number of elementary dipoles which are distributed on a triangle net of the cortical surface. During the inverse calculation, the minimum norm least square method was used.

$$\Delta^2 = \left\| Lj - m \right\|^2 + \lambda \left\| Wj \right\|^2 \quad (8)$$

where L is the lead field, j is the source, m is scalp-recorded signal, W is a matrix with depth information, and λ is the regularization parameter, which was determined using the L-curve criterion.

III. Results

Fig. 2 shows the equivalent dipole solutions of three trials for both left and right hand imagination reconstructed in the alpha band. Because in alpha band, a synchronization appears in ipsilateral side of the brain during motor imagination, the equivalent dipole solution represented well this phenomenon in the ipsilateral side.

Fig. 3 shows typical examples of the current density distribution of different trials during left and right hand imagination. It is seen that the CCD imaging represents the synchronization in the ipsilateral cortex.

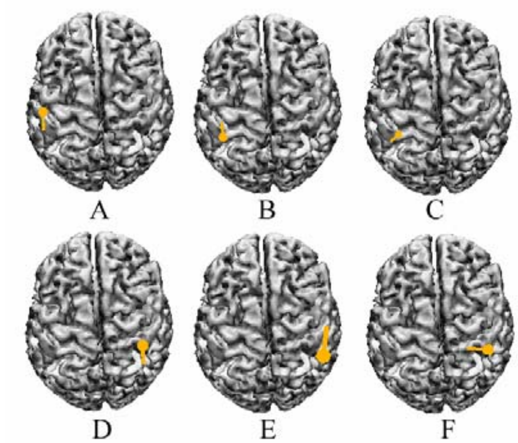

Fig. 2 Equivalent dipole solutions of different trials during imagination of left or right hand in alpha band. (A,B,C - left-hand imagery; D,E,F - right-hand imagery).

A simple classification rule was defined based on source localization results. For equivalent dipole analysis, if the equivalent dipole is located at the ipsilateral side in alpha band we judge it as correct, otherwise wrong. For CCD, the 15% largest current density is summarized, thus in the alpha band, the summed current density should be stronger at the ipsilateral side than at the contralateral side. Based on these criteria, we obtained 81.1% and 79.4% classification accuracy for the subject studied by means of

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the equivalent dipole analysis and CCD imaging respectively.

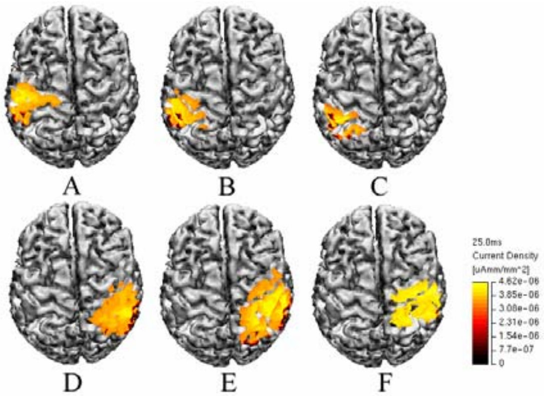

Fig. 3. Cortical current density distribution of different trials during motor imagination. (A,B,C - left-hand imagery; D,E,F - right-hand imagery).

IV. Discussion

The present study initially tested the hypothesis that source analysis methods such as dipole localization and cortical imaging can be applied to motor imagery tasks classification in a BCI system. The promising results we have obtained in this preliminary study suggest that the source analysis approach has the advantage of performing the classification directly without a training process. It can be directly used in the testing data if correct implementation and judging criterion are set. It may reflect physiologically what's happening in the brain during the motor imagery.

In the present study, two important tools were used: ICA is effective to extract the useful information; source estimation can be used to identify the imagined hand movements by revealing the origins of the brain activities so as to manifest the difference between mental tasks directly.

It is also obvious that the frequency band and time points are important for classification accuracy. How to choose the correct frequency and time is another issue that should be further explored.

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